

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JNANASANGAMA”, BELAGAVI -590018, KARNATAKA



A JavaScript and JQuery

Mini-Project Report

On

“Solar System Simulator”

*A Mini-project report submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Design Engineering** of Visvesvaraya Technological University, Belgaum.*

Submitted by:

Raghav Purushotham (1AT22CG068)

&

Rahul Mahato (1AT22CG069)

Under the Guidance of:

Ms. Niranjani M

(Asst. Prof. Dept. of CSE)



Department of Computer Science and Engineering

ATRIA INSTITUTE OF TECHNOLOGY BANGALORE- 560024

CERTIFICATE

Certified that the project work entitled “**Solar System Simulator**”, carried out by RAGHAV PURUSHOTHAM (1AT22CG068) & RAHUL MAHATO (1AT22CG069), and are bonafide students of Atria Institute of Technology, Bangalore, in partial fulfillment for the award of Bachelor of Engineering in Computer Science & Design Engineering of Visvesvaraya Technological University, Belgaum during the academic year 2023-2024. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library.

The project report has been approved as it satisfies requirement in respect of project work prescribed for the said degree.

Signature of Guide

Prof. Niranjani M
Asst. Prof ,Dept. of CSE

Signature of HOD

Dr. Devi Kannan
HOD, Dept. of CSE

DECLARATION

We, RAGHAV PURUSHOTHAM (1AT22CG068) & RAHUL MAHATO (1AT22CG069) students of IV semester B. E in Computer Science & Design Engineering at Atria Institute of Technology, hereby declare that the project work entitled “*Solar System Simulator*” has been carried out under the supervision of **Prof. Niranjani M**, Assistant Professor, Dept. of CS&E, Atria Institute of Technology and submitted in partial fulfillment of the course requirements for the award of degree in B.E in Computer Science & Design Engineering of Visvesvaraya Technological University, Belagavi during the year 2023-2024. We further declare that the report has not been submitted to any other University for the award of any other degree.

Place: Bangalore

Date:-

**Signature of the
Students**

NAME : RAGHAV PURUSHOTHAM

USN : 1AT22CG068

NAME : RAHUL MAHATO

USN : 1AT22CG069

ABSTRACT

Project Title: Solar System Viewer

This project presents an interactive 3D model of the solar system, designed to provide an engaging and educational experience for users interested in astronomy. Developed using HTML, CSS, and JavaScript, the model simulates the orbits of planets around the sun, complete with realistic animations and interactive controls.

The visual interface allows users to toggle between different views, zoom in and out, and adjust the scale to explore the solar system in detail. Additionally, an integrated audio component enhances the immersive experience with thematic background music that can be controlled directly from the interface.

This project aims to make learning about the solar system accessible and enjoyable, providing a dynamic tool for both educational and recreational purposes. The implementation demonstrates effective use of modern web technologies to create complex visualizations and interactive elements that respond seamlessly to user inputs.

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Raghav Purushotham (1AT22CG068)

Rahul Mahato (1AT22CG069)

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Introduction

Java Script:

JavaScript is a high-level, versatile programming language primarily used for enhancing web pages and building web applications. Initially created by Brendan Eich in 1995, JavaScript has evolved into one of the most widely-used languages in web development.

Core Features:

1. **Client-Side Scripting:** JavaScript runs in the browser, allowing developers to create interactive and dynamic web pages. It can manipulate HTML and CSS, respond to user actions, and control multimedia, making websites more engaging and functional.
2. **Versatility:** While it started as a client-side language, JavaScript has expanded beyond the browser. With the advent of Node.js, JavaScript is now used for server-side programming as well. This means developers can use JavaScript for both frontend and backend development, streamlining the development process.
3. **Event-Driven Programming:** JavaScript is built around the concept of events. It responds to user actions like clicks, form submissions, and keystrokes, enabling developers to create responsive and interactive user experiences.
4. **Dynamic Typing:** JavaScript is dynamically typed, meaning variables can hold values of any type and types can change at runtime. This flexibility allows for rapid development but can also introduce type-related bugs if not managed carefully.
5. **Rich Ecosystem:** JavaScript boasts a vast ecosystem of libraries and frameworks like React, Angular, and Vue.js, which simplify the development of complex applications. Additionally, npm (Node Package Manager) provides access to a huge repository of open-source packages and modules.
6. **Asynchronous Programming:** JavaScript supports asynchronous programming through callbacks, promises, and the newer async/await syntax. This allows for

non-blocking operations, such as making network requests or handling file I/O, improving performance and user experience.

Use Cases:

- **Web Development:** Enhancing user interfaces, handling form validations, and creating interactive features.
- **Server-Side Development:** Building scalable backend services and APIs with Node.js.
- **Mobile Development:** Creating cross-platform mobile apps with frameworks like React Native.
- **Game Development:** Building browser-based games using HTML5 and JavaScript.

Aim of the Project:

The primary aim of this project is to develop an interactive, web-based platform that allows users to explore the solar system in a visually engaging and educational manner. By creating a 3D model of the solar system, this project seeks to provide users with a realistic and immersive experience that enhances their understanding of planetary motion and celestial mechanics.

This project targets a broad audience, including students, educators, and astronomy enthusiasts, by offering a dynamic tool that can be used for both learning and entertainment. Through the use of modern web technologies such as HTML, CSS, and JavaScript, the model offers various interactive features that enable users to manipulate their view of the solar system, zoom in on specific planets, and adjust the scale to better comprehend the vastness and complexity of space.

In addition to visual simulations, the project integrates audio components to create a more immersive experience. Users can control background music that complements the visual exploration, adding an additional sensory dimension to the educational experience. This multisensory approach aims to engage users more fully, promoting a deeper curiosity and appreciation for the intricacies of our solar system.

Furthermore, this project demonstrates the potential of using web technologies to create sophisticated educational tools that are accessible to a wide audience. By showcasing the orbits of planets and their relative sizes and distances from the sun, the project provides valuable insights into the scale and dynamics of the solar system. Ultimately, this project aims to inspire a sense of wonder about space while providing a practical and enjoyable way to learn about astronomy.

Technical Implementation:

The technical implementation of this project involves the development of a 3D interactive solar system model using a combination of HTML, CSS, and JavaScript. This integration of technologies provides a seamless and responsive user experience, enabling real-time interaction with the solar system model.

1. **HTML Structure:**

The HTML file serves as the backbone of the project, defining the structure and content of the webpage. It includes elements representing the sun and each planet, with unique identifiers for styling and scripting purposes. The document structure supports dynamic interactions through integrated scripts and links to external resources such as fonts and libraries.

2. **CSS Styling:**

The CSS file is responsible for the visual styling of the solar system model, using modern design principles to create a realistic and aesthetically pleasing representation. Key features include:

- **3D Transformations:** CSS properties like `transform` and `transform-style` are used to simulate the three-dimensional appearance and motion of planets.
- **Animations:** Keyframe animations define the orbits of planets around the sun, giving them a continuous, realistic motion. The animations are designed to be smooth and responsive, using properties like `animation-name`, `animation-duration`, and `animation-timing-function`.
- **Visual Effects:** Shadows, gradients, and background colors enhance the visual appeal, creating a more immersive experience.

3. **JavaScript Interactivity:**

JavaScript plays a crucial role in making the solar system model interactive and responsive to user input. Key functionalities include:

- **Event Handling:** jQuery is used to handle events such as button clicks, allowing users to control the playback of background music and toggle UI elements.

- **Dynamic Updates:** Scripts update the content dynamically based on user interactions, such as changing the button text to reflect the current state of the music playback.
- **External Libraries:** The project incorporates external libraries like jQuery for simplified DOM manipulation and event handling, as well as Prefixfree for automatic vendor prefixing of CSS properties, ensuring cross-browser compatibility.

4. **Audio Integration:**

An HTML5 audio element is used to embed background music into the project, enhancing the user experience. JavaScript controls allow users to start and stop the music, adding an additional layer of interaction.

5. **Responsive Design:**

The project is designed to be responsive, ensuring compatibility across various devices and screen sizes. CSS media queries and flexible layouts ensure that the model scales appropriately, maintaining usability and aesthetic quality on both desktop and mobile platforms.

Through this technical implementation, the project achieves its aim of providing an engaging and educational tool for exploring the solar system. The combination of visual design, interactive elements, and responsive functionality demonstrates the potential of web technologies to create innovative educational experiences.

Requirements and Specifications

The development of the interactive 3D solar system model involves several technical and functional requirements to ensure a seamless user experience. This section outlines the necessary specifications and requirements needed for the successful implementation and operation of the project.

Functional Requirements

1. 3D Visualization:

- The system must render a three-dimensional representation of the solar system, displaying the sun and planets with accurate relative positions and orbits.
- Users should be able to switch between 2D and 3D views to enhance their understanding of planetary motion.

2. User Interaction:

- Users must be able to control the size and distance scaling of the planetary orbits via interactive controls.
- The interface should provide options to start and stop background music, with visual feedback reflecting the current state.

3. Planetary Information:

- Hovering over a planet should display basic information such as the planet's name and position in orbit.
- The interface should highlight the active planet when a user selects it for detailed viewing.

Technical Requirements

1. Web Technologies:

- **HTML5** is required to structure the webpage and integrate multimedia elements such as audio.
- **CSS3** is necessary for styling, animations, and 3D transformations.
- **JavaScript** and **jQuery** are required for dynamic content updates and user interaction handling.

2. Browser Compatibility:

- Ensure the application is compatible with major modern web browsers, including Chrome, Firefox, Safari, and Edge.

- Manually add vendor prefixes in CSS where necessary to maintain cross-browser compatibility for properties like transform and animation.

3. **Performance:**

- The system should load efficiently with minimal lag, optimizing performance for smooth animations and interactions.
- The use of asynchronous loading techniques, such as the `<script>` tag with `defer` or `async`, can improve page load times.

Design Specifications

1. **User Interface:**

- The design should be intuitive, with controls easily accessible to users.
- The visual layout must be clean and modern, with a dark theme to mimic the space environment.

2. **Responsive Design:**

- The application must be responsive, adjusting layouts and interactions for various screen sizes and orientations.
- CSS media queries should be utilized to optimize the display for both desktop and mobile devices.

3. **Audio Integration:**

- The system should preload audio files for seamless playback.
- Controls for audio must be responsive and provide clear indications of the current playback state.

Environmental Requirements

1. **Internet Connectivity:**

- While the basic functionality should operate offline, internet connectivity is required to access external libraries and fonts from CDNs.

2. **Hardware:**

- A device with moderate processing power is recommended to handle the animations and interactions smoothly.

3. **Operating Systems:**

- The application should be platform-independent, functioning

across various operating systems such as Windows, macOS, and Linux.

By meeting these requirements and specifications, the project ensures a robust and engaging educational tool that effectively leverages modern web technologies to simulate the solar system.

Source Code:

-Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>3D Solar System</title>
  <link rel="stylesheet" href="style.css">
  <script
src="https://cdnjs.cloudflare.com/ajax/libs/prefixfree/1.0.7/prefixfree.min.js">
</script>
  <script
src="https://ajax.googleapis.com/ajax/libs/jquery/2.1.3/jquery.min.js"></
script>

  <script>$(document).ready(function(){
    let flipFlop = false;

    function changeButtonText() {
      let button = $("#startStopButton");
      if (flipFlop === false){
        button.text("Stop Music");
        flipFlop = true;
      } else {
        button.text("Start Music")
        flipFlop = false;
      }
    }
    $("#startStopButton").click(function(){
      changeButtonText();
    })
  });</script>

  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Major+Mono+Display&display=swap" rel="stylesheet">
```

```

</head>
<body class="opening hide-UI view-2D zoom-large data-close controls-
close">
<audio id="solarSound" src="solar-system-music.mp3" preload="auto"
autoplay loop></audio>
<div id="navbar">
  <a id="toggle-data" href="#data"><i class="icon-data"></i>Data</a>
  <a id="toggle-controls" href="#controls"><i
class="icon-controls"></i>Controls</a>
</div>
<div id="data">
  <a class="sun" title="sun" href="#sunspeed">sun</a>
  <a class="mercury" title="mercury" href="#mercuryspeed">mercury</a>
  <a class="venus" title="venus" href="#venusspeed">venus</a>
  <a class="earth active" title="earth" href="#earthspeed">earth</a>
  <a class="mars" title="mars" href="#marsspeed">mars</a>
  <a class="jupiter" title="jupiter" href="#jupiterspeed">jupiter</a>
  <a class="saturn" title="saturn" href="#saturnspeed">saturn</a>
  <a class="uranus" title="uranus" href="#uranusspeed">uranus</a>
  <a class="neptune" title="neptune" href="#neptunespeed">neptune</a>
</div>
<div id="controls">
  <label class="set-view">
    <input type="checkbox">
  </label>
  <label class="set-zoom">
    <input type="checkbox">
  </label>
  <label>
    <input type="radio" class="set-speed" name="scale" checked>
    <span>speed</span>
  </label>
  <label>
    <input type="radio" class="set-size" name="scale">
    <span>size</span>
  </label>
  <label>
    <input type="radio" class="set-distance" name="scale">
    <span>distance</span>
  </label>
  <label>

```



```
<button id="startStopButton">Start Music</button>
</label>
</div>
<div id="universe" class="scale-stretched">
  <div id="galaxy">
    <div id="solar-system" class="earth">
      <div id="mercury" class="orbit">
        <div class="pos">
          <div class="planet">
            <dl class="infos">
              <dt>mercury</dt>
              <dd><span></span></dd>
            </dl>
          </div>
        </div>
      </div>
    </div>
    <div id="venus" class="orbit">
      <div class="pos">
        <div class="planet">
          <dl class="infos">
            <dt>venus</dt>
            <dd><span></span></dd>
          </dl>
        </div>
      </div>
    </div>
    <div id="earth" class="orbit">
      <div class="pos">
        <div class="orbit">
          <div class="pos">
            <div class="moon"></div>
          </div>
        </div>
        <div class="planet">
          <dl class="infos">
            <dt>earth</dt>
            <dd><span></span></dd>
          </dl>
        </div>
      </div>
    </div>
  </div>
</div>
```

```
<div id="mars" class="orbit">
  <div class="pos">
    <div class="planet">
      <dl class="infos">
        <dt>mars</dt>
        <dd><span></span></dd>
      </dl>
    </div>
  </div>
</div>
<div id="jupiter" class="orbit">
  <div class="pos">
    <div class="planet">
      <dl class="infos">
        <dt>jupiter</dt>
        <dd><span></span></dd>
      </dl>
    </div>
  </div>
</div>
<div id="saturn" class="orbit">
  <div class="pos">
    <div class="planet">
      <div class="ring"></div>
      <dl class="infos">
        <dt>saturn</dt>
        <dd><span></span></dd>
      </dl>
    </div>
  </div>
</div>
<div id="uranus" class="orbit">
  <div class="pos">
    <div class="planet">
      <dl class="infos">
        <dt>uranus</dt>
        <dd><span></span></dd>
      </dl>
    </div>
  </div>
</div>
```

```
<div id="neptune" class="orbit">
  <div class="pos">
    <div class="planet">
      <dl class="infos">
        <dt>neptune</dt>
        <dd><span></span></dd>
      </dl>
    </div>
  </div>
</div>
<div id="sun">
  <dl class="infos">
    <dt>sun</dt>
    <dd><span></span></dd>
  </dl>
</div>
</div>
</div>
</body>
<script rel="script" src="script.js"></script>
</body>
</html>
```

- script.js

```
$(window).load(function(){

    var body = $("body"),
        universe = $("#universe"),
        solarsys = $("#solar-system");

    var init = function() {
        body.removeClass('view-2D opening').addClass("view-3D").delay(2000).queue(function() {
            $(this).removeClass('hide-UI').addClass("set-speed");
            $(this).dequeue();
        });
    };

    var setView = function(view) { universe.removeClass().addClass(view); };

    $("#toggle-data").click(function(e) {
        body.toggleClass("data-open data-close");
        e.preventDefault();
    });

    $("#toggle-controls").click(function(e) {
        body.toggleClass("controls-open controls-close");
        e.preventDefault();
    });

    $("#data a").click(function(e) {
        var ref = $(this).attr("class");
        solarsys.removeClass().addClass(ref);
        $(this).parent().find('a').removeClass('active');
        $(this).addClass('active');
        e.preventDefault();
    });

    $(".set-view").click(function() { body.toggleClass("view-3D view-2D"); });
    $(".set-zoom").click(function() { body.toggleClass("zoom-large zoom-close"); });
    $(".set-speed").click(function() { setView("scale-stretched set-speed"); });
    $(".set-size").click(function() { setView("scale-s set-size"); });
    $(".set-distance").click(function() { setView("scale-d set-distance"); });
});
```

```
init();
$(document).ready(function () {
    var solarSound = document.getElementById('solarSound');
    var isPlaying = false;

    // Play the audio when the "Start/Stop" button is clicked
    $("#startStopButton").click(function () {
        if (isPlaying) {
            solarSound.pause(); // Pause the audio
            isPlaying = false;
        } else {
            solarSound.currentTime = 0; // Reset the audio to the beginning
            solarSound.play(); // Play the audio
            isPlaying = true;
        }
    });
});
```

- Styles.css:

```
@import url(https://fonts.googleapis.com/css?family=Open+Sans:400,300);
```

```
html, body, div, span, applet, object, iframe, h1, h2, h3, h4, h5, h6, p,  
blockquote, pre, a, abbr, acronym, address, big, cite, code, del, dfn, em, img,  
ins, kbd, q, s, samp, small, strike, strong, sub, sup, tt, var, b, u, i, center, dl, dt,  
dd, ol, ul, li, fieldset, form, label, legend, table, caption, tbody, tfoot, thead, tr,  
th, td, article, aside, canvas, details, embed, figure, figcaption, footer, header,  
hgroup, menu, nav, output, ruby, section, summary, time, mark, audio, video  
{
```

```
    border: 0;  
    font-size: 100%;  
    font: inherit;  
    vertical-align: baseline;  
    margin: 0;  
    padding: 0;
```

```
}
```

```
article, aside, details, figcaption, figure, footer, header, hgroup, menu, nav,  
section {  
    display: block;  
}
```

```
body {  
    line-height: 1;  
}
```

```
ol, ul {  
    list-style: none;  
}
```

```
blockquote, q {  
    quotes: none;  
}
```

```
blockquote:before, blockquote:after, q:before, q:after {  
    content: none;  
}
```

```
table {
```

```
border-collapse: collapse;
border-spacing: 0;
}

* {
  -moz-box-sizing: border-box;
  box-sizing: border-box;
  box-sizing: border-box;
}

body {
  font-size: 10px;
  font-family: 'Major Mono Display', sans-serif;
  font-weight: 300;
  background-color: #08090A;
}

#universe {
  z-index: 1;
  position: absolute;
  overflow: hidden;
  width: 100%;
  height: 100%;
  background-position: center 40%;
  background-repeat: no-repeat;
  background-size: cover;
}

#galaxy {
  position: relative;
  width: 100%;
  height: 100%;
}

#solar-system {
  position: absolute;
  width: 100%;
  height: 100%;
  transform-style: preserve-3d;
}
```

```
.orbit {  
  position: absolute;  
  top: 50%;  
  left: 50%;  
  border: 1px solid rgba(255, 255, 255, 0.2);  
  border-radius: 50%;  
  transform-style: preserve-3d;  
  animation-name: orbit;  
  animation-iteration-count: infinite;  
  animation-timing-function: linear;  
}
```

```
.orbit .orbit {  
  animation-name: suborbit;  
}
```

```
.pos {  
  position: absolute;  
  top: 50%;  
  width: 2em;  
  height: 2em;  
  margin-top: -1em;  
  margin-left: -1em;  
  transform-style: preserve-3d;  
  animation-name: invert;  
  animation-iteration-count: infinite;  
  animation-timing-function: linear;  
}
```

```
#sun, .planet, #earth .moon {  
  position: absolute;  
  top: 50%;  
  left: 50%;  
  width: 1em;  
  height: 1em;  
  margin-top: -0.5em;  
  margin-left: -0.5em;  
  border-radius: 50%;  
  transform-style: preserve-3d;  
}
```



```
#sun {  
  background-color: #FB7209;  
  background-repeat: no-repeat;  
  background-size: cover;  
  box-shadow: 0 0 60px rgba(255, 160, 60, 0.4);  
}
```

```
.planet {  
  background-color: #202020;  
  background-repeat: no-repeat;  
  background-size: cover;  
  animation-iteration-count: infinite;  
  animation-timing-function: linear;  
}
```

```
.moon {  
  background-color: #202020;  
  background-repeat: no-repeat;  
  background-size: cover;  
  animation-iteration-count: infinite;  
  animation-timing-function: linear;  
}
```

```
.ring {  
  position: absolute;  
  top: 50%;  
  left: 50%;  
  border-radius: 50%;  
}
```

```
#saturn .ring {  
  width: 2em;  
  height: 2em;  
  margin-top: -1em;  
  margin-left: -1em;  
  border: 0.3em solid rgba(160, 147, 130, 0.7);  
  animation-iteration-count: infinite;  
  animation-timing-function: linear;  
}
```

```
#saturn .ring:after {
```

```
border-radius: 50%;
position: absolute;
content: "";
top: 50%;
left: 50%;
width: 2.2em;
height: 2.2em;
margin-top: -1.1em;
margin-left: -1.1em;
border: 0.07em solid rgba(160, 147, 130, 0.5);
box-sizing: border-box;
box-sizing: border-box;
}

/* ----- planets
index */
#mercury {
    z-index: 10;
}

#venus {
    z-index: 9;
}

#earth {
    z-index: 8;
}

#moon {
    z-index: 7;
}

#mars {
    z-index: 6;
}

#jupiter {
    z-index: 5;
}

#saturn {
```

```
    z-index: 4;
}

#uranus {
    z-index: 3;
}

#neptune {
    z-index: 2;
}

#sun {
    z-index: 1;
}

/* ----- animations
*/
@keyframes orbit {
    0% {
        transform: rotateZ(0deg);
    }

    100% {
        transform: rotateZ(-360deg);
    }
}

@keyframes suborbit {
    0% {
        transform: rotateX(90deg) rotateZ(0deg);
    }

    100% {
        transform: rotateX(90deg) rotateZ(-360deg);
    }
}

@keyframes invert {
    0% {
        transform: rotateX(-90deg) rotateY(360deg) rotateZ(0deg);
    }
}
```

```
100% {
  transform: rotateX(-90deg) rotateY(0deg) rotateZ(0deg);
}
}

/* ----- opening */
.view-3D .opening #sun,
.view-3D .opening .orbit,
.view-3D .opening .pos,
.view-3D .opening .planet,
.view-3D .opening .satelite,
.view-3D .opening .ring {
  transition-duration: 4s;
}

#universe.opening #sun {
  box-shadow: 0 0 0 rgba(255, 160, 60, 0);
}

/* ----- 2D view */
.view-2D.zoom-large #sun {
  transform-style: flat;
}

.view-2D.zoom-large .orbit {
  transform-style: flat;
}

.view-2D #sun,
.view-2D .ring {
  transform: rotateX(0deg);
}

.view-2D .planet,
.view-2D .moon {
  transform: rotateX(90deg);
}

/* ----- 3D view */
.view-3D.zoom-large #sun {
```

```
    transform-style: preserve-3d;
}

.view-3D.zoom-large .orbit {
    transform-style: preserve-3d;
}

.view-3D #solar-system {
    transform: rotateX(75deg);
}

.view-3D #sun {
    transform: rotateX(-90deg);
}

.view-3D .ring {
    transform: rotateX(90deg);
}

.view-3D .planet,
.view-3D .moon {
    transform: rotateX(0deg);
}

/* ----- large zoom
*/
.zoom-large #solar-system {
    width: 100%;
}

.zoom-large.view-2D .scale-stretched #solar-system {
    font-size: 26%;
}

.zoom-large.view-3D .scale-stretched #solar-system {
    font-size: 62%;
}

.zoom-large.view-2D .scale-d #solar-system {
    font-size: 22%;
}
```

```
.zoom-large.view-3D .scale-d #solar-system {
    font-size: 48%;
}

.zoom-large.view-2D .scale-s #solar-system {
    font-size: 7%;
}

.zoom-large.view-3D .scale-s #solar-system {
    font-size: 14%;
}

/* ----- close zoom
*/

.zoom-close #solar-system {
    width: 200%;
    font-size: 100%;
}

.zoom-close .scale-stretched #solar-system {
    margin-left: -100%;
}

.zoom-close .scale-d #solar-system {
    margin-left: -106%;
}

.zoom-close.view-2D .scale-d #solar-system {
    font-size: 84%;
}

.zoom-close.view-3D .scale-d #solar-system {
    font-size: 84%;
}

.zoom-close .scale-s #solar-system {
    margin-left: -145%;
}

.zoom-close.view-2D .scale-s #solar-system {
```

```

    font-size: 40%;
}

.zoom-close.view-3D .scale-s #solar-system {
    font-size: 40%;
}

.zoom-close .orbit,
.zoom-close .pos,
.zoom-close .planet,
.zoom-close .satellite,
.zoom-close .ring {
    animation-play-state: paused !important;
    animation-duration: 0s !important;
    animation: none !important;
}

.zoom-close.view-2D .planet,
.zoom-close.view-2D .moon {
    transform: rotateX(0deg);
}

.zoom-close.view-3D .planet {
    transform: rotateX(-90deg);
}

.zoom-close.view-3D .moon {
    transform: rotateX(90deg);
}

.zoom-close .pos {
    left: 100% !important;
    top: 50% !important;
}

/* ----- data */
/* ----- speed */
/* sideral years */
#mercury .pos,
#mercury .planet,
#mercury.orbit {

```

```
    animation-duration: 2.89016s;
}

#venus .pos,
#venus .planet,
#venus.orbit {
    animation-duration: 7.38237s;
}

#earth .pos,
#earth .planet,
#earth.orbit {
    animation-duration: 12.00021s;
}

#earth .orbit .pos,
#earth .orbit {
    animation-duration: 0.89764s;
}

#mars .pos,
#mars .planet,
#mars.orbit {
    animation-duration: 22.57017s;
}

#jupiter .pos,
#jupiter .planet,
#jupiter.orbit {
    animation-duration: 142.35138s;
}

#saturn .pos,
#saturn .planet,
#saturn.orbit,
#saturn .ring {
    animation-duration: 353.36998s;
}

#uranus .pos,
#uranus .planet,
```



```
#uranus.orbit {  
  animation-duration: 1008.20215s;  
}
```

```
#neptune .pos,  
#neptune .planet,  
#neptune.orbit {  
  animation-duration: 1977.49584s;  
}
```

```
/* ----- planets sizes  
*/  
/* ----- stretched  
sizes */
```

```
.scale-stretched #sun {  
  font-size: 24em;  
}
```

```
.scale-stretched #mercury .planet {  
  font-size: 1.5em;  
}
```

```
.scale-stretched #venus .planet {  
  font-size: 3.72em;  
}
```

```
.scale-stretched #earth .planet {  
  font-size: 3.92em;  
}
```

```
.scale-stretched #earth .moon {  
  font-size: 1.2em;  
}
```

```
.scale-stretched #mars .planet {  
  font-size: 2.9em;  
}
```

```
.scale-stretched #jupiter .planet {  
  font-size: 12em;  
}
```

```
.scale-stretched #saturn .planet {  
    font-size: 10.8em;  
}
```

```
.scale-stretched #uranus .planet {  
    font-size: 4.68em;  
}
```

```
.scale-stretched #neptune .planet {  
    font-size: 4.9em;  
}
```

```
/* ----- scaled sizes  
*/
```

```
/*  
(planets radius * 2) * ratio  
*/
```

```
/*$distanceScaleRatio: 0.000075em;*/
```

```
/* ----- planets  
distance scale sizes */
```

```
.scale-d #sun {  
    font-size: 41.73048em;  
}
```

```
.scale-d #mercury .planet {  
    font-size: 0.14634em;  
}
```

```
.scale-d #venus .planet {  
    font-size: 0.36306em;  
}
```

```
.scale-d #earth .planet {  
    font-size: 0.38226em;  
}
```

```
.scale-d #earth .moon {  
    font-size: 0.08226em;  
}
```

```
.scale-d #mars .planet {  
  font-size: 0.20334em;  
}
```

```
.scale-d #jupiter .planet {  
  font-size: 4.19466em;  
}
```

```
.scale-d #saturn .planet {  
  font-size: 3.49392em;  
}
```

```
.scale-d #uranus .planet {  
  font-size: 1.52172em;  
}
```

```
.scale-d #neptune .planet {  
  font-size: 1.47732em;  
}
```

```
/* ----- planet size  
scale sizes */
```

```
.scale-s #sun {  
  font-size: 417.3048em;  
}
```

```
.scale-s #mercury .planet {  
  font-size: 1.4634em;  
}
```

```
.scale-s #venus .planet {  
  font-size: 3.6306em;  
}
```

```
.scale-s #earth .planet {  
  font-size: 3.8226em;  
}
```

```
.scale-s #earth .moon {  
  font-size: 0.8226em;  
}
```

```
.scale-s #mars .planet {  
    font-size: 2.0334em;  
}
```

```
.scale-s #jupiter .planet {  
    font-size: 41.9466em;  
}
```

```
.scale-s #saturn .planet {  
    font-size: 34.9392em;  
}
```

```
.scale-s #uranus .planet {  
    font-size: 15.2172em;  
}
```

```
.scale-s #neptune .planet {  
    font-size: 14.7732em;  
}
```

```
/* ----- stretched  
orbits */
```

```
.scale-stretched #mercury.orbit {  
    width: 32em;  
    height: 32em;  
    margin-top: -16em;  
    margin-left: -16em;  
}
```

```
.scale-stretched #venus.orbit {  
    width: 40em;  
    height: 40em;  
    margin-top: -20em;  
    margin-left: -20em;  
}
```

```
.scale-stretched #earth.orbit {  
    width: 56em;  
    height: 56em;  
    margin-top: -28em;
```

```
margin-left: -28em;
}

.scale-stretched #earth .orbit {
width: 6em;
height: 6em;
margin-top: -3em;
margin-left: -3em;
}

.scale-stretched #mars.orbit {
width: 72em;
height: 72em;
margin-top: -36em;
margin-left: -36em;
}

.scale-stretched #jupiter.orbit {
width: 100em;
height: 100em;
margin-top: -50em;
margin-left: -50em;
}

.scale-stretched #saturn.orbit {
width: 150em;
height: 150em;
margin-top: -75em;
margin-left: -75em;
}

.scale-stretched #uranus.orbit {
width: 186em;
height: 186em;
margin-top: -93em;
margin-left: -93em;
}

.scale-stretched #neptune.orbit {
width: 210em;
height: 210em;
```

```
margin-top: -105em;
margin-left: -105em;
}

/* ----- scaled orbits
*/
/* ----- distance
scale orbits */
.scale-d #mercury.orbit {
  width: 44.74176em;
  height: 44.74176em;
  margin-top: -22.37088em;
  margin-left: -22.37088em;
}

.scale-d #venus.orbit {
  width: 47.35737em;
  height: 47.35737em;
  margin-top: -23.67869em;
  margin-left: -23.67869em;
}

.scale-d #earth.orbit {
  width: 49.50959em;
  height: 49.50959em;
  margin-top: -24.75479em;
  margin-left: -24.75479em;
}

.scale-d #mars.orbit {
  width: 53.58356em;
  height: 53.58356em;
  margin-top: -26.79178em;
  margin-left: -26.79178em;
}

.scale-d #jupiter.orbit {
  width: 82.2042em;
  height: 82.2042em;
  margin-top: -41.1021em;
  margin-left: -41.1021em;
```

```

}

.scale-d #saturn.orbit {
  width: 115.91713em;
  height: 115.91713em;
  margin-top: -57.95857em;
  margin-left: -57.95857em;
}

.scale-d #uranus.orbit {
  width: 191.00471em;
  height: 191.00471em;
  margin-top: -95.50235em;
  margin-left: -95.50235em;
}

.scale-d #neptune.orbit {
  width: 275.64709em;
  height: 275.64709em;
  margin-top: -137.82355em;
  margin-left: -137.82355em;
}

/* Moon */
.scale-d #earth .orbit {
  width: 1em;
  height: 1em;
  margin-top: -0.5em;
  margin-left: -0.5em;
}

/* ----- size scale
orbits */
.scale-s #mercury.orbit {
  width: 441.3048em;
  height: 441.3048em;
  margin-top: -220.6524em;
  margin-left: -220.6524em;
}

.scale-s #venus.orbit {

```

```
width: 457.3048em;  
height: 457.3048em;  
margin-top: -228.6524em;  
margin-left: -228.6524em;  
}
```

```
.scale-s #earth.orbit {  
width: 473.3048em;  
height: 473.3048em;  
margin-top: -236.6524em;  
margin-left: -236.6524em;  
}
```

```
.scale-s #mars.orbit {  
width: 489.3048em;  
height: 489.3048em;  
margin-top: -244.6524em;  
margin-left: -244.6524em;  
}
```

```
.scale-s #jupiter.orbit {  
width: 561.3048em;  
height: 561.3048em;  
margin-top: -280.6524em;  
margin-left: -280.6524em;  
}
```

```
.scale-s #saturn.orbit {  
width: 705.3048em;  
height: 705.3048em;  
margin-top: -352.6524em;  
margin-left: -352.6524em;  
}
```

```
.scale-s #uranus.orbit {  
width: 817.3048em;  
height: 817.3048em;  
margin-top: -408.6524em;  
margin-left: -408.6524em;  
}
```



```

.scale-s #neptune.orbit {
  width: 881.3048em;
  height: 881.3048em;
  margin-top: -440.6524em;
  margin-left: -440.6524em;
}

/* Moon */
.scale-s #earth .orbit {
  width: 1em;
  height: 1em;
  margin-top: -0.5em;
  margin-left: -0.5em;
}

/* ----- text infos
data */
/* ----- speed */
.set-speed dl.infos dd span:after {
  content: 'orbit velocity';
}

.set-speed #sun dl.infos dd:after {
  content: '0 km/h';
}

.set-speed #mercury dl.infos dd:after {
  content: '170,503 km/h';
}

.set-speed #venus dl.infos dd:after {
  content: '126,074 km/h';
}

.set-speed #earth dl.infos dd:after {
  content: '107,218 km/h';
}

.set-speed #mars dl.infos dd:after {
  content: '86,677 km/h';
}

```

```
.set-speed #jupiter dl.infos dd:after {
    content: '47,002 km/h';
}

.set-speed #saturn dl.infos dd:after {
    content: '34,701 km/h';
}

.set-speed #uranus dl.infos dd:after {
    content: '24,477 km/h';
}

.set-speed #neptune dl.infos dd:after {
    content: '19,566 km/h';
}

/* ----- size */
.set-size dl.infos dd span:after {
    content: 'Equatorial Circumference';
}

.set-size #sun dl.infos dd:after {
    content: '4,370,005 km';
}

.set-size #mercury dl.infos dd:after {
    content: '15,329 km';
}

.set-size #venus dl.infos dd:after {
    content: '38,024 km';
}

.set-size #earth dl.infos dd:after {
    content: '40,030 km';
}

.set-size #mars dl.infos dd:after {
    content: '21,296 km';
}
```

```
.set-size #jupiter dl.infos dd:after {  
    content: '439,263 km';  
}
```

```
.set-size #saturn dl.infos dd:after {  
    content: '365,882 km';  
}
```

```
.set-size #uranus dl.infos dd:after {  
    content: '159,354 km';  
}
```

```
.set-size #neptune dl.infos dd:after {  
    content: '154,704 km';  
}
```

```
/* ----- distance */
```

```
.set-distance dl.infos dd span:after {  
    content: 'From Sun';  
}
```

```
.set-distance #sun dl.infos dd span:after {  
    content: 'From Earth';  
}
```

```
.set-distance #sun dl.infos dd:after {  
    content: '149,598,262 km';  
}
```

```
.set-distance #mercury dl.infos dd:after {  
    content: '57,909,227 km';  
}
```

```
.set-distance #venus dl.infos dd:after {  
    content: '108,209,475 km';  
}
```

```
.set-distance #earth dl.infos dd:after {  
    content: '149,598,262 km';  
}
```

```

.set-distance #mars dl.infos dd:after {
    content: '227,943,824 km';
}

.set-distance #jupiter dl.infos dd:after {
    content: '778,340,821 km';
}

.set-distance #saturn dl.infos dd:after {
    content: '1,426,666,422 km';
}

.set-distance #uranus dl.infos dd:after {
    content: '2,870,658,186 km';
}

.set-distance #neptune dl.infos dd:after {
    content: '4,498,396,441 km';
}

/* ----- planets
starting position and lighting effect */
/* ----- mercury ;
pos: Top */
#mercury .pos {
    left: 50%;
    top: -1%;
}

#mercury .planet {
    animation-name: shadow-mercury;
}

@keyframes shadow-mercury {
    0% {
        box-shadow: inset 0 0 1px rgba(0, 0, 0, 0.5);
        /* TOP */
    }

    25% {

```

```

    box-shadow: inset 4px 0 2px rgba(0, 0, 0, 0.5);
    /* LEFT */
}

50% {
    box-shadow: inset 13px -5px 4px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

50.01% {
    box-shadow: inset -13px -5px 4px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

75% {
    box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

100% {
    box-shadow: inset 0 0 1px rgba(0, 0, 0, 0.5);
    /* TOP */
}
}

.scaled.view-2D #mercury .planet,
.scaled.view-3D #mercury .planet {
    box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

/* ----- venus ; pos:
Left */
#venus .pos {
    left: 0;
    top: 50%;
}

#venus .planet {
    animation-name: shadow-venus;
}

```

```

@keyframes shadow-venus {
  0% {
    box-shadow: inset 4px 0 2px rgba(0, 0, 0, 0.5);
    /* LEFT */
  }

  25% {
    box-shadow: inset 22px -20px 10px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
  }

  25.5% {
    box-shadow: inset -22px -20px 10px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
  }

  50% {
    box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
    /* RIGHT */
  }

  75% {
    box-shadow: inset 0 0 1px rgba(0, 0, 0, 0.5);
    /* TOP */
  }

  100% {
    box-shadow: inset 4px 0 2px rgba(0, 0, 0, 0.5);
    /* LEFT */
  }
}

.scaled.view-2D #venus .planet,
.scaled.view-3D #venus .planet {
  box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
  /* RIGHT */
}

/* ----- earth ; pos:
Right */

```

```
#earth .pos {
  left: 100%;
  top: 50%;
}

#earth .planet {
  animation-name: shadow-earth;
}

@keyframes shadow-earth {
  0% {
    box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
    /* RIGHT */
  }

  25% {
    box-shadow: inset 0 0 1px rgba(0, 0, 0, 0.5);
    /* TOP */
  }

  50% {
    box-shadow: inset 4px 0 2px rgba(0, 0, 0, 0.5);
    /* LEFT */
  }

  75% {
    box-shadow: inset 24px -20px 15px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
  }

  75.01% {
    box-shadow: inset -24px -20px 15px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
  }

  100% {
    box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
    /* RIGHT */
  }
}
```

```

.scaled.view-2D #earth .planet,
.scaled.view-3D #earth .planet {
    box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

/* ----- moon */
#earth .orbit .pos {
    left: 100%;
    top: 50%;
}

/* ----- mars ; pos:
Bottom */
#mars .pos {
    left: 50%;
    top: 100%;
}

#mars .planet {
    animation-name: shadow-mars;
}

@keyframes shadow-mars {
    0% {
        box-shadow: inset -18px -10px 10px rgba(0, 0, 0, 0.5);
        /* BOTTOM */
    }

    25% {
        box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
        /* RIGHT */
    }

    50% {
        box-shadow: inset 0 1px 1px rgba(0, 0, 0, 0.5);
        /* TOP */
    }

    75% {
        box-shadow: inset 4px 0 2px rgba(0, 0, 0, 0.5);
    }
}

```



```

    /* LEFT */
}

99.99% {
    box-shadow: inset 18px -10px 10px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

100% {
    box-shadow: inset -18px -10px 10px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}
}

.scaled.view-2D #mars .planet,
.scaled.view-3D #mars .planet {
    box-shadow: inset -4px 0 2px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

/* ----- jupiter ; pos:
Right */
#jupiter .pos {
    left: 100%;
    top: 50%;
}

#jupiter .planet {
    animation-name: shadow-jupiter;
}

@keyframes shadow-jupiter {
    0% {
        box-shadow: inset -16px 0 5px rgba(0, 0, 0, 0.5);
        /* RIGHT */
    }

    25% {
        box-shadow: inset 0 3px 2px rgba(0, 0, 0, 0.5);
        /* TOP */
    }
}

```

```

50% {
  box-shadow: inset 16px 0 5px rgba(0, 0, 0, 0.5);
  /* LEFT */
}

75% {
  box-shadow: inset 94px -30px 40px rgba(0, 0, 0, 0.5);
  /* BOTTOM */
}

75.01% {
  box-shadow: inset -94px -30px 40px rgba(0, 0, 0, 0.5);
  /* BOTTOM */
}

100% {
  box-shadow: inset -16px 0 5px rgba(0, 0, 0, 0.5);
  /* RIGHT */
}
}

.scaled.view-2D #jupiter .planet,
.scaled.view-3D #jupiter .planet {
  box-shadow: inset -16px 0 5px rgba(0, 0, 0, 0.5);
  /* RIGHT */
}

/* ----- saturn ; pos:
Left */
#saturn .pos {
  left: 0%;
  top: 50%;
}

#saturn .planet {
  animation-name: shadow-saturn;
}

@keyframes shadow-saturn {
  0% {

```

```

    box-shadow: inset 16px 0 5px rgba(0, 0, 0, 0.5);
    /* LEFT */
}

25% {
    box-shadow: inset 80px -30px 50px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

25.01% {
    box-shadow: inset -94px -30px 40px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

50% {
    box-shadow: inset -16px 0 5px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

75% {
    box-shadow: inset -2px 3px 2px rgba(0, 0, 0, 0.5);
    /* TOP */
}

100% {
    box-shadow: inset 16px 0 5px rgba(0, 0, 0, 0.5);
    /* LEFT */
}
}

.scaled.view-2D #saturn .planet,
.scaled.view-3D #saturn .planet {
    box-shadow: inset -16px 0 5px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

/* ----- saturn ring
*/

#saturn .ring { animation-name: shadow-saturn-ring; }

```

```

@keyframes
shadow-saturn-ring{ 0% { box-shadow: inset -16px 0 4px rgba(20, 40,
50, 0.45); opacity: .4; }
25% { box-shadow: inset -5px -30px 12px rgba(20, 40, 50,
0.3); opacity: 1; }
25.01% { box-shadow: inset 5px -30px 12px rgba(20, 40, 50,
0.3); opacity: 1; }
50% { box-shadow: inset 16px 0 4px rgba(20, 40, 50,
0.45); opacity: .4; }
75% { box-shadow: inset 0 30px 50px rgba(20, 40, 50, 0);
opacity: .3; }
100% { box-shadow: inset -16px 0 4px rgba(20, 40, 50,
0.45); opacity: .4; }
}

```

```

.scaled.view-2D #saturn .ring,
.scaled.view-3D #saturn .ring { box-shadow: inset 16px 0 4px rgba(20,
40, 50, 0.6); }

```

```

/* ----- uranus ; pos:
Left */

```

```

#uranus .pos {
left: 0;
top: 50%;
}

```

```

#uranus .planet {
animation-name: shadow-uranus;
}

```

```

@keyframes shadow-uranus {
0% {
box-shadow: inset 8px 0 5px rgba(0, 0, 0, 0.5);
/* LEFT */
}

25% {
box-shadow: inset 40px -15px 40px rgba(0, 0, 0, 0.5);
/* BOTTOM */
}
}

```

```

}

25.01% {
    box-shadow: inset -40px -15px 40px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

50% {
    box-shadow: inset -8px 0 5px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

75% {
    box-shadow: inset 0 0 2px rgba(0, 0, 0, 0.5);
    /* TOP */
}

100% {
    box-shadow: inset 8px 0 5px rgba(0, 0, 0, 0.5);
    /* LEFT */
}
}

.scaled.view-2D #uranus .planet,
.scaled.view-3D #uranus .planet {
    box-shadow: inset -8px 0 5px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

/* ----- neptune ;
pos: Top */
#neptune .pos {
    left: 50%;
    top: 0;
}

#neptune .planet {
    animation-name: shadow-neptune;
}

@keyframes shadow-neptune {

```

```

0% {
    box-shadow: inset 0 0 2px rgba(0, 0, 0, 0.5);
    /* TOP */
}

25% {
    box-shadow: inset 12px 0 5px rgba(0, 0, 0, 0.5);
    /* LEFT */
}

50% {
    box-shadow: inset 50px -15px 40px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

50.01% {
    box-shadow: inset -50px -15px 40px rgba(0, 0, 0, 0.5);
    /* BOTTOM */
}

75% {
    box-shadow: inset -12px 0 5px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

100% {
    box-shadow: inset 0 0 2px rgba(0, 0, 0, 0.5);
    /* TOP */
}
}

.scaled.view-2D #neptune .planet,
.scaled.view-3D #neptune .planet {
    box-shadow: inset -12px 0 5px rgba(0, 0, 0, 0.5);
    /* RIGHT */
}

/* ----- DEBUG
.pos,
.planet,
.orbit,

```

```
.ring { animation-duration: 20s!important; }
```

```
/* */
```

```
/* ----- navigation  
styles */
```

```
dl.infos {  
    position: absolute;  
    display: block;  
    opacity: 0;  
    width: 100%;  
    height: 100%;  
    margin-top: -90%;  
    margin-left: 90%;  
    padding-left: 100%;  
    transform-origin: 100% 100%;  
    transform-style: preserve-3d;  
    transform: rotateX(90deg);  
}
```

```
dl.infos:before {  
    position: absolute;  
    content: "";  
    width: 15px;  
    height: 30px;  
    left: 15px;  
    bottom: 0;  
    border-top: 1px solid white;  
    border-left: 1px solid white;  
    transform-style: preserve-3d;  
    transform: skew(-45deg, 0deg);  
    box-shadow: inset 1px 1px black;  
}
```

```
dl.infos dt {  
    position: absolute;  
    left: 50px;  
    margin-bottom: 26px;  
    bottom: 30px;  
    color: #FFF;  
    font-size: 14px;  
    text-shadow: 1px 1px 2px black;
```

```
}
```

```
dl.infos dd:after {  
    position: absolute;  
    left: 50px;  
    bottom: 30px;  
    width: 300px;  
    color: #FFF;  
    font-size: 22px;  
    text-shadow: 1px 1px 2px black;  
}
```

```
dl.infos dd span:after {  
    position: absolute;  
    left: 50px;  
    bottom: 14px;  
    width: 300px;  
    color: #FFF;  
    font-size: 11px;  
    text-shadow: 1px 1px 2px black;  
}
```

```
.sun #sun .infos,  
.mercury #mercury .infos,  
.venus #venus .infos,  
.earth #earth .infos,  
.mars #mars .infos,  
.jupiter #jupiter .infos,  
.saturn #saturn .infos,  
.uranus #uranus .infos,  
.neptune #neptune .infos {  
    display: block;  
    opacity: 1;  
    transform: rotateX(0deg);  
}
```

```
.mercury #mercury.orbit,  
.venus #venus.orbit,  
.earth #earth.orbit,  
.mars #mars.orbit,  
.jupiter #jupiter.orbit,
```



```
.saturn #saturn.orbit,  
.uranus #uranus.orbit,  
.neptune #neptune.orbit {  
    border: 1px solid rgba(255, 255, 255, 0.8);  
}  
  
.hide-UI h1,  
.hide-UI #data,  
.hide-UI dl.infos,  
.hide-UI #controls {  
    opacity: 0 !important;  
    margin-top: -30px;  
}  
  
.hide-UI #data {  
    margin-bottom: -30px;  
}  
  
.hide-UI .orbit {  
    border: 1px solid rgba(255, 255, 255, 0.2) !important;  
}  
  
h1 {  
    width: 100%;  
    font-weight: 600;  
    font-size: 14px;  
    text-align: center;  
    color: rgba(255, 255, 255, 0.8);  
}  
  
h1 span, #navbar h1 a {  
    display: inline;  
    position: relative;  
    padding: 0;  
    font-weight: 300;  
    font-size: 14px;  
    text-align: center;  
    color: rgba(255, 255, 255, 0.5);  
}  
  
#navbar, #controls, #data {
```

```
background: rgba(0, 0, 0, 0.4);
}

#navbar {
  z-index: 99;
  position: absolute;
  top: 0;
  left: 0;
  padding: 16px;
  width: 100%;
  height: 48px;
}

#navbar a, #data a, #controls label {
  color: rgba(255, 255, 255, 0.6);
  display: block;
  position: relative;
  text-decoration: none;
}

#navbar a:hover, #data a:hover, #controls label:hover {
  color: #FFF;
}

#data a.active {
  color: #0CF;
}

#navbar a {
  position: absolute;
  top: 0;
  height: 48px;
  padding: 16px;
  font-size: 14px;
}

#toggle-data {
  left: 0;
}

#toggle-controls {
```

```
    right: 0;
}

#data, #controls {
    z-index: 99;
    position: fixed;
    opacity: 1;
    top: 49px;
    padding: 16px;
}

.data-close #data {
    left: -100%;
}

.data-open #data {
    left: 0px;
}

.controls-close #controls {
    right: -100%;
}

.controls-open #controls {
    right: 0px;
}

#data a {
    margin-bottom: 1px;
    padding: 6px 10px;
    font-size: 18px;
}

#controls label {
    opacity: .6;
    height: 24px;
    margin-bottom: 28px;
}

#controls label:hover {
    opacity: 1;
```

```
}
```

```
#controls label:before {  
    position: absolute;  
    display: block;  
    width: 20px;  
    height: 20px;  
    margin-top: 3px;  
    text-align: center;  
    color: #FFF;  
    z-index: 99;  
}
```

```
#controls label span {  
    display: block;  
    margin-left: 36px;  
    padding-top: 4px;  
    font-size: 18px;  
    color: #FFF;  
}
```

```
#controls input {  
    display: block;  
    appearance: none;  
}
```

```
#controls input[type="radio"]:before {  
    content: " "  
    display: block;  
    position: absolute;  
    width: 22px;  
    height: 22px;  
    border: 2px solid #FFF;  
    border-radius: 16px;  
}
```

```
#controls input:checked[type="radio"]:after {  
    content: " "  
    display: block;  
    top: 3px;  
    margin-left: 3px;
```

```
position: absolute;
width: 20px;
height: 20px;
border-radius: 10px;
background: #FFF;
z-index: 99;
}

#controls input[type="checkbox"]:before {
  content: "";
  display: block;
  position: absolute;
  width: 100px;
  height: 22px;
  border: 2px solid #FFF;
  border-radius: 16px;
}

#controls input[type="checkbox"]:after {
  content: "";
  display: block;
  top: 3px;
  position: absolute;
  width: 20px;
  height: 20px;
  border-radius: 10px;
  background: #FFF;
  z-index: 99;
}

#controls label.set-view:before {
  font: bold small-caps 11px/20px sans-serif;
}

.view-3D #controls label.set-view:before {
  content: '2D';
  margin-left: 82px;
}

.view-2D #controls label.set-view:before {
  content: '3D';
```

```
margin-left: 7px;
}

.view-3D #controls .set-view input:after {
margin-left: 3px;
}

.view-2D #controls .set-view input:after {
margin-left: 81px;
}

#controls label.set-zoom:before {
font: normal small-caps 18px/14px sans-serif;
}

.zoom-large #controls label.set-zoom:before {
content: '+';
margin-left: 82px;
}

.zoom-close #controls label.set-zoom:before {
content: '-';
margin-left: 7px;
}

.zoom-large #controls .set-zoom input:after {
margin-left: 3px;
}

.zoom-close #controls .set-zoom input:after {
margin-left: 81px;
}

/* ----- transitions
*/

.pos {
transition-property: top, left;
}

#solar-system, .orbit,
.planet,
```

```
.satelite,  
.ring {  
    transition-property: width, height, top, left, margin-left, margin-top, webkit-  
transform;  
}
```

```
#sun, .icon {  
    transition-property: width, height, webkit-transform;  
}
```

```
#solar-system,  
#sun,  
.orbit,  
.pos,  
.planet,  
.satelite,  
.ring,  
.infos,  
.icon {  
    transition-duration: .8s;  
    transition-timing-function: ease-in-out;  
}
```

```
#solar-system,  
#sun,  
.planet,  
.satelite,  
.ring {  
    transition-delay: 0s;  
}
```

```
.pos {  
    transition-delay: 1s;  
}
```

```
.opening #solar-system,  
.opening #sun,  
.opening .orbit,  
.opening .pos,  
.opening .planet,  
.opening .satelite,
```

```
.opening .ring,  
.opening .infos,  
.opening h1,  
.opening #data,  
.opening dl.infos,  
.opening #controls {  
    transition-duration: 0s;  
    transition-delay: 0s;  
}  
  
.opening .pos {  
    transition-delay: 0s;  
}  
  
h1, #data, #controls {  
    transition-property: opacity, margin;  
    transition-duration: .8s;  
    transition-timing-function: ease-in-out;  
}  
  
h1 {  
    transition-delay: .35s;  
}  
  
#data {  
    transition-delay: .7s;  
}  
  
#controls {  
    transition-delay: 1s;  
}  
  
/* ----- images */  
body {  
    background-image: url(IMGES/Background_image.png);  
}  
  
#universe {  
    background-image: url(IMGES/universe.png);  
}
```



```
#sun {  
    background-image: url(IMAGEs/sun.png);  
}  
  
#mercury .planet {  
    background-image: url(IMAGEs/mercury.png);  
}  
  
#venus .planet {  
    background-image: url(IMAGEs/venus.png);  
}  
  
#earth .planet {  
    background-image: url(IMAGEs/earth.png);  
}  
  
#earth .moon {  
    background-image: url(IMAGEs/moon.png);  
}  
  
#mars .planet {  
    background-image: url(IMAGEs/mars.png);  
}  
  
#jupiter .planet {  
    background-image: url(IMAGEs/jupiter.png);  
}  
  
#saturn .planet {  
    background-image: url(IMAGEs/saturn.png);  
}  
  
#uranus .planet {  
    background-image: url(IMAGEs/uranus.png);  
}  
  
#neptune .planet {  
    background-image: url(IMAGEs/neptune.png);  
}
```

```
/* ----- responsive
styles */
@media screen and (max-width: 299px) {
  #universe {
    font-size: 20%;
  }
}

@media screen and (min-width: 300px) {
  #universe {
    font-size: 24%;
  }
}

@media screen and (min-width: 500px) {
  #universe {
    font-size: 36%;
  }
}

@media screen and (min-width: 600px) {
  #universe {
    font-size: 44%;
  }
}

@media screen and (min-width: 760px) {
  #universe {
    font-size: 58%;
  }
}

@media screen and (min-width: 1000px) {
  #universe {
    font-size: 73%;
  }

  #navbar, #controls, #data {
    background: transparent;
  }
}
```

```
#navbar a {
  display: none;
}

h1 {
  font-size: 22px;
  margin-top: 8px;
}

#controls {
  padding-right: 32px;
  top: 12px;
}

#data {
  position: fixed;
  top: inherit;
  bottom: 0;
  width: 100%;
  text-align: center;
}

#data a {
  display: inline-block;
  text-align: center;
  font-size: 20px;
  padding: 15px 15px;
}

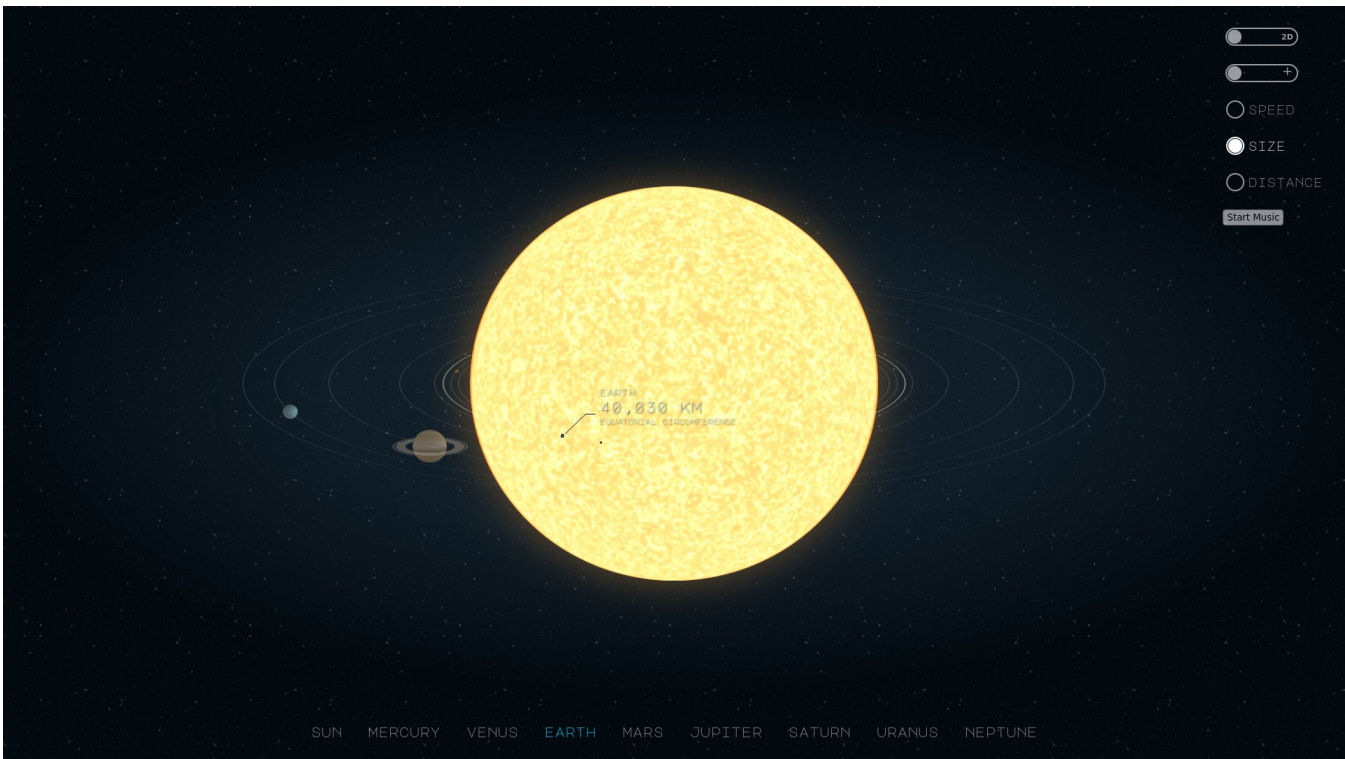
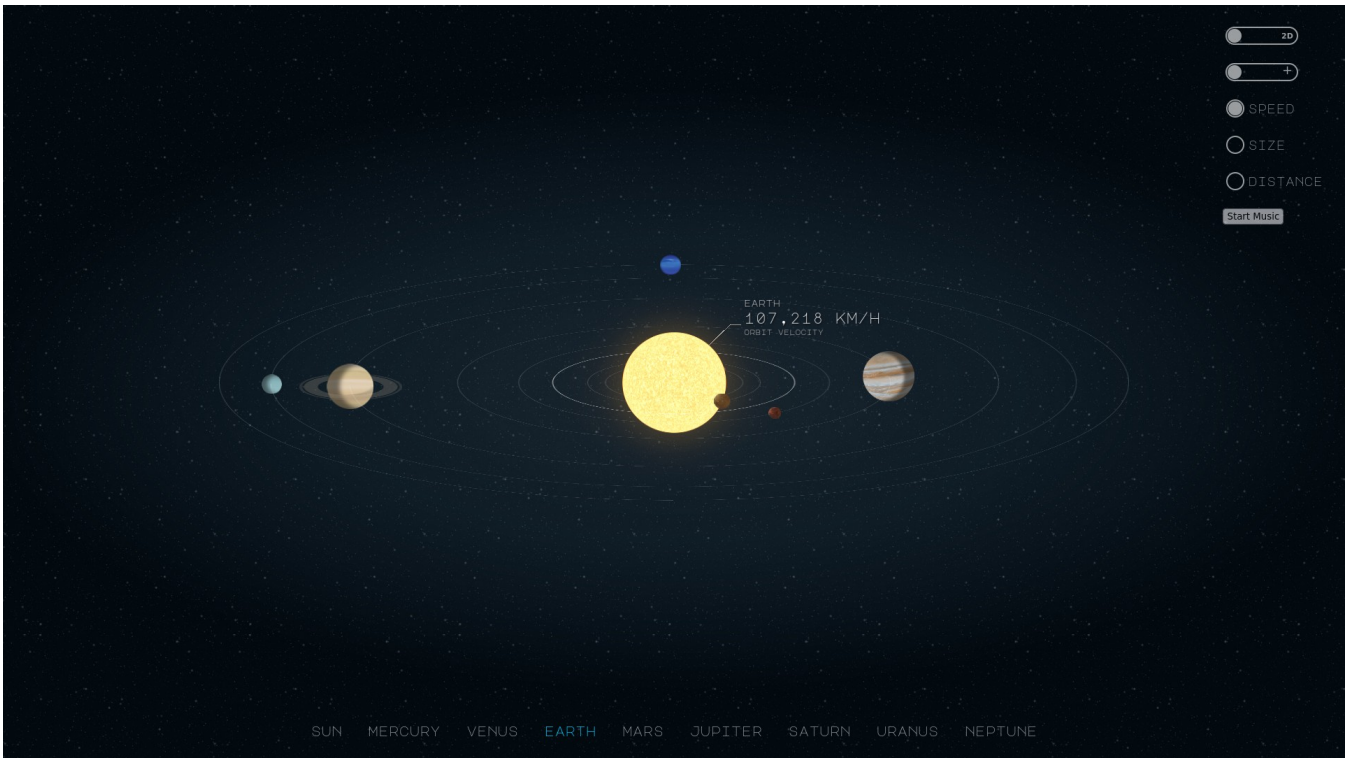
.data-open #data, .data-close #data {
  left: 0px;
}

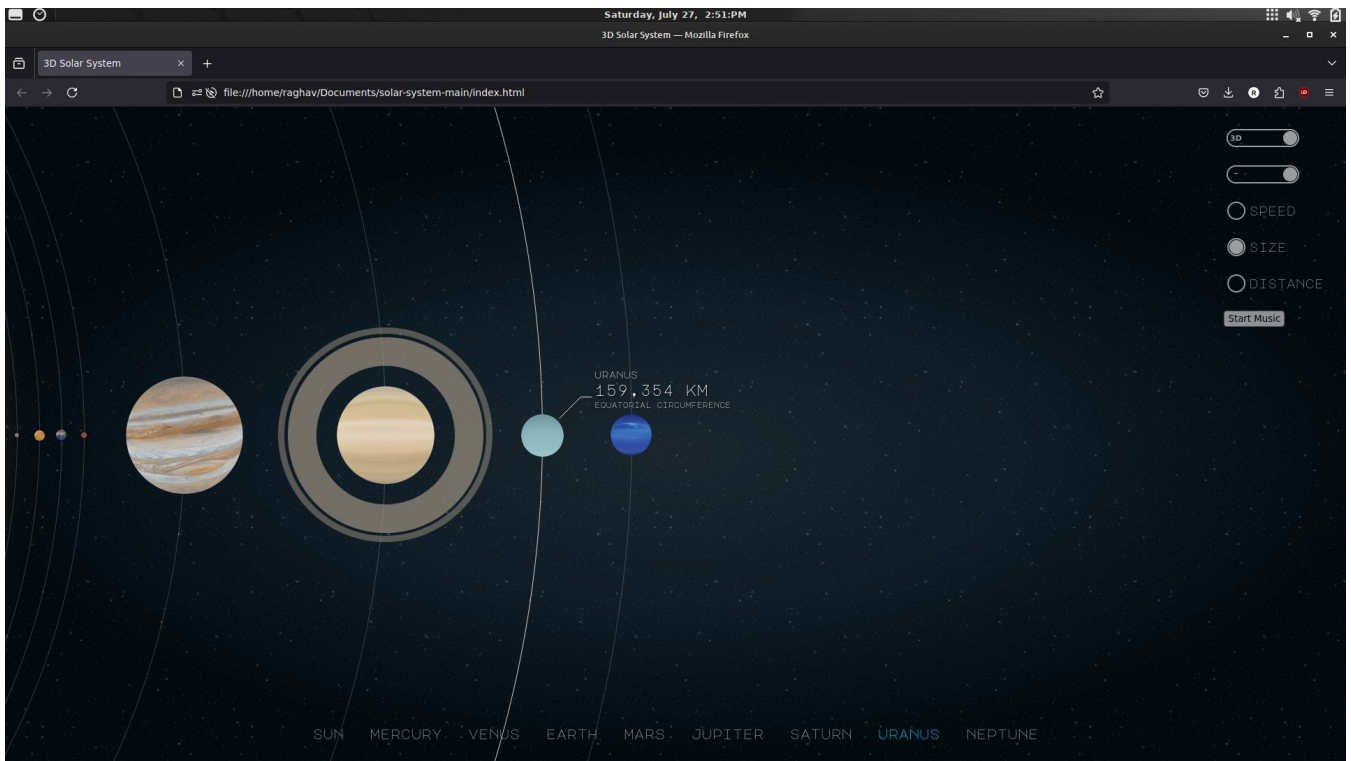
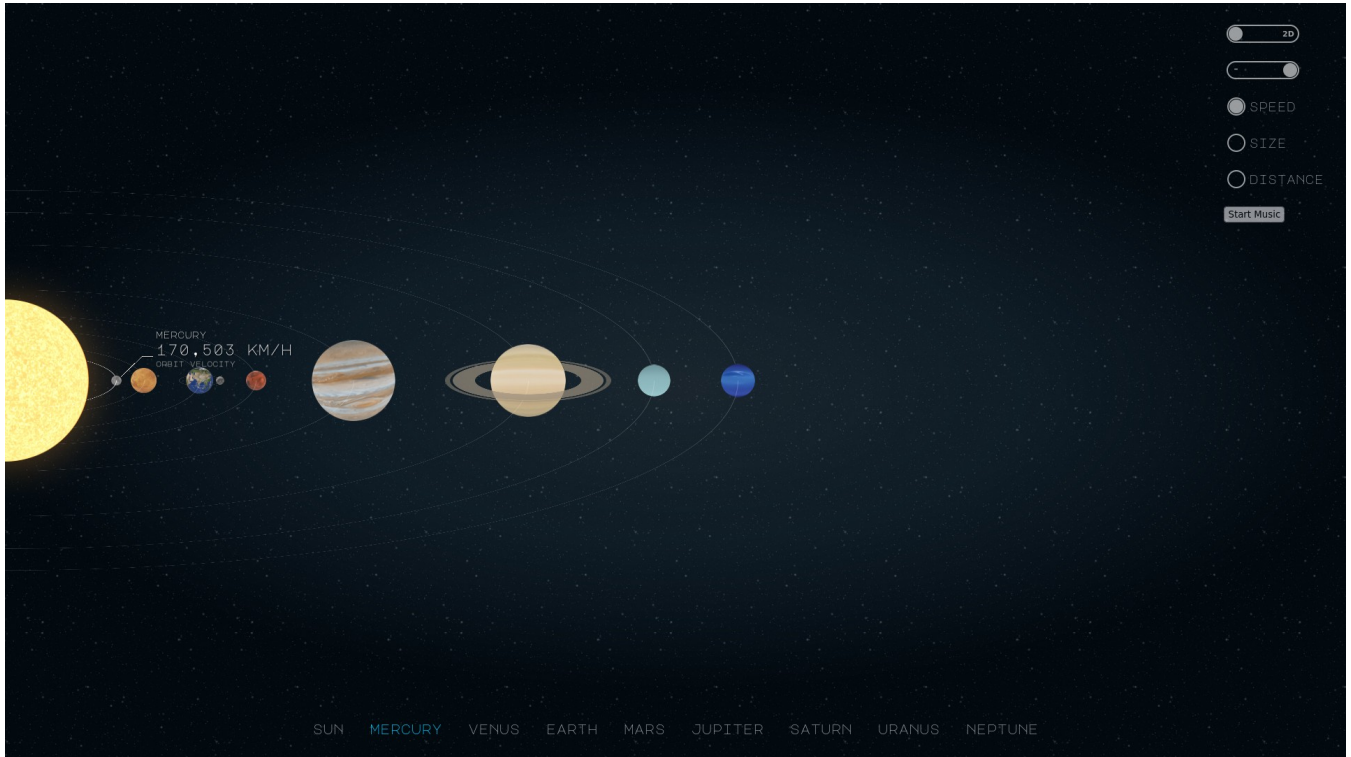
.controls-open #controls, .controls-close #controls {
  right: 0px;
}
}

@media screen and (min-width: 1300px) {
  #universe {
    font-size: 100%;
  }
}
```

```
}  
}  
  
@media screen and (min-width: 1600px) {  
  .zoom-close .scale-d #solar-system {  
    margin-left: -120%;  
  }  
}
```

Screenshots:





Conclusion:

This project successfully demonstrates the power of modern web technologies in creating an interactive and educational platform for exploring the solar system. Through the development of a 3D model, users are provided with an immersive experience that enhances their understanding of astronomical concepts and planetary dynamics.

The integration of HTML, CSS, and JavaScript enables the creation of a responsive and visually appealing interface, allowing users to engage with the solar system in a meaningful way. By incorporating interactive features such as zooming, scaling, and orbit speed adjustments, the project effectively engages users of all ages and promotes curiosity about space.

The project's focus on cross-browser compatibility and responsive design ensures that the model is accessible to a wide audience, regardless of their device or browser preferences. By carefully optimizing performance and leveraging external libraries, the project achieves smooth animations and interactions, providing an enjoyable user experience.

Additionally, the audio integration adds an extra sensory dimension to the model, enhancing the overall atmosphere and making the exploration more captivating. The ability to toggle music playback allows users to customize their experience according to their preferences.

Overall, this project illustrates the potential of combining technology and education to create tools that are both informative and entertaining. It serves as a valuable resource for students, educators, and astronomy enthusiasts alike, offering a dynamic and engaging way to learn about the solar system. As web technologies continue to evolve, projects like this one can further inspire innovative approaches to education and interactive content delivery.

References:

1. <https://stackoverflow.com/questions/73227097/how-to-create-a-simple-orbit-animation-using-3d-css>
2. <https://www.geeksforgeeks.org/foundation-css-orbit/>
3. <https://www.javatpoint.com/javascript-tutorial>